

Andrés Riveros Valdevenito

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 in andres-riveros-valdevenito 🏷 Andrés Riveros Valdevenito 📄 Andrés Riveros Valdevenito

Education

PhD	Columbia University , PhD in Statistics. • Co-advised by Professor Marcel Nutz & Professor Philip Protter.	Sep 2020 - Expected May 2026
M Phil	Columbia University , Master of Philosophy in Statistics. • As part of the PhD program.	Sep 2020 - May 2024
MA	Columbia University , Master of Arts in Statistics. • As part of the PhD program.	Sep 2020 - Feb 2023
MS	Facultad de Ciencias Físicas y Matemáticas (FCFM), Universidad de Chile , Master of Engineering Sciences, minor in Applied Mathematics. • With Maximum Distinction. • Advised by Professor Jaime San Martín.	Mar 2014 - Jun 2020
Prof. Deg.	Facultad de Ciencias Físicas y Matemáticas (FCFM), Universidad de Chile , Mathematical Engineer. • With Maximum Distinction.	Mar 2014 - Jun 2020
BS	Facultad de Ciencias Físicas y Matemáticas (FCFM), Universidad de Chile , Engineering Sciences in Mathematics. • With Distinction.	Mar 2014 - Dec 2018

Teaching

Teaching Assistant at Columbia University :	Sep 2020 - Present
• Stochastic Methods in Finance STAT GR4265 (Fall 2025), • Probability Qualifying Exam Review Session Teacher (Summer 2023, 2024, 2025), • PhD Probability II STAT GR6302 (Spring 2024, Spring 2025), • PhD Probability I STAT GR6301 (Fall 2023, Fall 2024), • Core Competency Exam Review Session Teacher (Spring 2024), • Probability STAT S4203/W5203 (Summer 2023), • Introduction to Statistics with Calculus STAT S1201/W1201 (Summer 2022, Spring 2023), • Linear Regression Models STAT S4205/S5205 (Fall 2022), • Introduction to Statistics STAT W1101 (Spring 2022, Fall & Spring 2021), • Introduction to Statistical Reasoning STAT W1001 (Summer 2021, Fall 2020).	
Teaching Assistant at Universidad de Chile, FCFM:	Mar 2014 - Jun 2020
• Supervisor of Entire Core Courses of Mathematics (2019,2020,2021), • Stochastic Calculus MA5402 (Spring 2018, 2019), • Markov Processes MA4401 (Fall 2018), • Introduction to Calculus MA1001 (Fall 2017), • Linear Algebra MA1102 (Spring 2016): Grader & Test Supervisor, • Introduction to Calculus MA1001 (Fall 2016): Grader & Test Supervisor.	

Honours & Awards

Columbia University:

Sep 2020

- Dean's Fellow, Graduate School of Arts & Sciences.

ANID (CONICYT):

Mar 2019

- Excellence Scholarship for National Science Masters Degree Students, Chile.

Experience

Dept. of Applied Mathematics, Universidad de Granada, Visiting Researcher.

Granada, Spain

- Invited by Professor Alejandro Gárriz.
- Ongoing project related to optimal transport theory.

June 2025

Center of Mathematical Modelling, Universidad de Chile, Research Intern.

Santiago, Chile

- Under guidance of Professor Joaquín Fontbona.
- Research review of state-of-the-art HMC methods.

Jan 2019

Superintendency of Banks and Financial Institutions (SBIF, now Financial Market Commission, CMF), Analytics Intern.

Santiago, Chile

- Under Patricio Jaramillo, director of the area of modelling and methodology.

Jan 2018

Assarlab, Faculty of Medicine, Universidad de Chile, Research Intern.

Santiago, Chile

- Under professor Rodrigo Assar.
- Created algorithms based on previous research to analyze the learning process of fish.

Jan 2017

Publications

Linear Convergence of Gradient Descent for Quadratically Regularized Optimal Transport.

2025+

A. González-Sanz, M. Nutz, **A. Riveros Valdevenito**,

Preprint, [arXiv:2509.08547](#) .

Monotonicity in Quadratically Regularized Linear Programs.

2025

A. González-Sanz, M. Nutz, **A. Riveros Valdevenito**,

SIAM Journal on Optimization, Vol 35, N°2, pp 1419-1437, [arXiv:2408.07871](#) .

Markov Jump Times and their Cox Construction.

2024

P. Protter, **A. Riveros Valdevenito**,

Annals of Operations Research, [arXiv:2310.07003](#) .

On the Guyon-Lekeufack Volatility Model.

2024

M. Nutz, **A. Riveros Valdevenito**,

Finance & Stochastics, Vol 28, N°4, pp 1203-1223, [arXiv:2307.01319](#) .

Propiedad de Martingala, Volatilidad Estocástica & Burbujas Financieras

2020

A. Riveros Valdevenito,

Master's Thesis, advised by professor Jaime San Martín, available at [respositorio uchile](#) .

Invited & Contributing Talks

- *IV Workshop on Stochastic Analysis and Optimization, applied to economics, finance, energy and insurance*, Puerto Natales, Chile, December 2025.
- *24th Northeastern Probability Seminar*, CUNY, November 2025.

- *9th Eastern Conference in Mathematical Finance* (Poster), CMU, October 2025.
- *Probability Seminar*, FCFM, Universidad de Chile, August 2025.
- *Young Researchers Seminar*, Universidad de Granada, June 2025.
- *Minghui Conference*, Columbia University, April 2025.
- *Statistics and Optimal Transport Workshop* (Poster), Columbia University, March 2025.
- *Statistics Student Seminar*, Columbia University, January 2025.
- *23rd Northeastern Probability Seminar*, Columbia University, November 2024.
- *Minghui Conference*, Columbia University, April 2024.
- *22nd Northeastern Probability Seminar*, NYU, November 2023.
- *Minghui Conference*, Columbia University, April 2023.
- *Oxford Statistics Student Seminar*, June 2020.

Referee

- Finance & Stochastics.

Technologies & Skills

Basic Programming: Matlab, SciLab, Python, R, Java, Unix, SQL, Visual Basic, Latex, MS Office.

Mathematical Programs: AMPL, BOCOP, FreeFem, STATA, Stan.

Languages: Fluent in English and Spanish. Basic French.